

Electrospinning, mechanical properties and cell behavior study of chitosan/PVA nanofibers

Mojtaba Koosha¹, Hamid Mirzadeh^{1,*}

¹Department of Polymer Engineering and Color Technology, Amirkabir University of Technology (Tehran Polytechnic), 424 Hafez Avenue, Tehran, Iran

mkoosha@aut.ac.ir, mirzadeh@aut.ac.ir

* Corresponding author.

E-mail address: mirzadeh@aut.ac.ir

ABSTRACT

Electrospinning process has been widely used to produce nanofibers from polymer blends. Poly (vinyl alcohol) (PVA) and chitosan (CS) have numerous biomedical applications such as wound healing and tissue engineering. Nanofibers of CS/PVA have been prepared by many works, however, a complete physico-chemical and mechanical characterization as well as cell behavior has not been reported. In this study, PVA and CS/PVA blend solutions in acetic acid 70% with different volume ratios (30/70, 50/50 and 70/30) were electrospun in constant electrospinning process parameters. The structure and morphology of nanofibrous mats were characterized by SEM, FTIR and XRD methods. The best nanofibrous mat was achieved from the CS/PVA 30/70 blend solution regarding the electrospinning throughput. The dynamic mechanical thermal analysis (DMTA) of PVA and CS/PVA 30/70 nanofibrous mats were measured which were not considered in the previous studies. DMTA results in accordance to the DSC analysis, approved the partial compatibility between the two polymers, while a single glass transition temperature was not observed for the blend. The tensile strength of PVA and CS/PVA nanofibers were also reported. Results of cell behavior study indicated that the heat stabilized nanofibrous mat CS/PVA 30/70 was able to support the attachment and proliferation of the fibroblast cells.

KEYWORDS

Chitosan; poly (vinyl alcohol); nanofibers; electrospinning; cell behavior;

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1. INTRODUCTION

Electrospinning is one of the best techniques for production of nanofibers from polymers, composites and ceramics¹. The nanofibrous structures show surprising characteristics such as high surface area to volume ratio and high porosity². In the electrospinning process, there are a number of factors which affect the morphology of the electrospun nanofibers. These are mainly classified into: (1) solution properties, such as concentration, viscosity, electric conductivity and surface tension and (2) process parameters, especially applied voltage, tip to collector distance (TCD), injection rate and environment temperature and humidity. These parameters determine the spinnability and final arrangement of nanofibers³.

Chitosan (CS), the N-deacetylated form of chitin, is a semi-natural biopolymer having invaluable biological properties such as hydrophilicity, biocompatibility, biodegradability and absorbable biodegradation products. CS is used in forms of hydrogels, films, microspheres and nanofibers. Nanofibrous mats of CS are of great interest in scaffolding materials because they mimic the structure and function of the natural extracellular matrix (ECM)^{4,5}. Nanofibers have the potential to provide an alternate artificial extracellular matrix for cell attachment and proliferation⁶. In recent years, researchers have prepared nonwoven nanofibrous mats of CS for tissue engineering⁷, membrane applications^{8,9} and wound dressings^{10,11}. However, CS is hardly electrospinnable¹². The main reason is that CS is only soluble in acidic solvents such as formic or acetic acid. NH_2 group in the saccharidic backbone of CS protonates into NH_3^+ in acidic media and makes it a cationic polyelectrolyte. In an electric field, large repulsive forces are developed between the positively charged groups which causes the incapability of CS for continuous fiber formation and produces non-uniform beaded morphology¹³. Strong hydrogen bonding between NH_2 and OH groups also plays an important role in the physical properties of CS solutions¹³. At low concentrations the polymer chains are not sufficient to form the fiber. By increasing the polymer concentration strong hydrogen bonding between the chains, makes a highly viscous 3D network structure which is difficult to electrospin. To solve this problem, two methods were utilized by researchers to overcome the difficulties in electrospinning of CS, which have led to ultrafine CS nanofibers with unbeaded morphology: (1) Blending of CS with a guest easily electrospinnable polymer such as poly (vinyl alcohol) (PVA)¹⁴⁻¹⁶, poly(ethylene oxide) (PEO)

¹⁷, etc. (2) reduction of repulsive forces between CS chains using environmentally harmful solvents such as trifluoroacetic acid^{18,19}. Although nanofibers of pure CS can be prepared by the latter method, utilization of harmful solvents are not favored, especially in bio-applications. In this regard, CS nanofibers are chiefly prepared by blending with PVA, because of its favorable interactions in the molecular level with CS^{16,20}.

Although CS/PVA nanofibers are prepared for a variety of applications, few papers are available on the complete physicochemical and mechanical characterization of the mats. Though, in this study, CS/PVA solutions were prepared in glacial acetic acid/water mixture (70/30 v/v) and blended in different ratios. The nanofibrous mats were prepared by the electrospinning method. The mats were characterized by scanning electron microscopy (SEM), differential scanning calorimetry (DSC) and X-Ray diffraction (XRD) methods and the effects of electrospinning process on the microstructure of CS/PVA nanofibers were discussed. Mechanical properties of CS/PVA nanofibrous mats were previously reported in the literature^{7,21} while dynamic mechanical properties, which can provide valuable data on phase transitions inside materials, were not considered for CS/PVA nanofibers. The best nanofibrous mat obtained in this work was selected and its tensile strength and dynamic mechanical properties were measured and compared to the PVA mat.

To approve the biocompatibility and cell attachment on the mat, the L-929 fibroblast cell behavior on the heat stabilized CS/PVA 30/70 nanofibrous mat, was also studied. The mat was stabilized physically by thermal treatment, since the use of chemical crosslinking agents may leave behind toxic, elutable compounds affecting the biocompatibility of the mats²².

2. MATERIALS AND METHODS

2.1 Materials

CS (viscosity of 1% solution in acetic acid 1% 50 mPa.s), degree of deacetylation 88% determined by FT-IR) was purchased from Sigma. PVA with molecular weight of about 89000-98000 g/mol and 99% degree of hydrolysis was provided by Sigma. Mixture of glacial acetic acid (AA) (Merck) and double distilled water (DW) in 70/30 volume ratio were used as solvent for preparation of electrospinning solutions. RPMI 1640, fetal bovine serum,

penicillin/streptomycin, L-glutamine, phosphate buffer saline (PBS) were provided by GIBCO company. Additionally, dimethyl sulfoxide (DMSO), ethanol, glutaraldehyde were obtained from Merck Company.

2.2 Solutions preparation

CS was dissolved in AA/DW (70/30) mixture by magnetic stirring overnight to prepare 7 wt% solutions. PVA solutions (7 wt%) were prepared by dissolving PVA powder in AA/DW (70/30) while the temperature was raised to 80°C for 3 hours to enhance dissolution of PVA crystals and the stirring of the solution was continued overnight. CS and PVA solutions were mixed in order to prepare solutions with CS/PVA volume ratios of 0/100, 30/70, 50/50 and 70/30 and 100/0. As the concentration of PVA and CS solutions were equal, the total concentration remained unchanged in all mixing ratios. The mats electrospun from the solutions were named as PVA, CS/PVA 30/70, CS/PVA 50/50, CS/PVA 70/30 and CS/PVA 100/0.

2.3 Viscometry

Viscosities of the solutions were measured with a Brookfield RVDV II+Pro extra (Brookfield Engineering Labs, Stoughton, MA). The viscosities were measured with the appropriate spindle. All viscosities were measured in rotational speed of 60 rpm.

2.4 Electrospinning

The solutions were electrospun immediately after preparation. The setup used for electrospinning (Electroris, Fanavaran Nano-Meghyas (FNM) Company, Tehran, Iran) is presented schematically in figure 1. A glass syringe (5 ml) was filled with the solution and placed in the holder which was moved by the pump to produce a constant injection rate. The solutions flowed from the syringe to a silicone tube connector, passed through the blunt-tipped stainless steel needle (gauge 18, inner diameter 0.84 mm) and entered the electric field which was provided by an adjustable high voltage supply. The needle was attached to a moving holder which can sweep in front of the mandrel. The device was provided with exhausting fans to evaporate the solvents. The remaining polymer chains forming the nanofibrous mats were collected on the aluminum foil covered on the rotating mandrel. The mandrel rotated with a tunable speed. All process parameters such as TCD,

injection rate, rotation speed of mandrel, scan position of the needle and the electric field were adjusted and controlled by a digital screen.

Electrospinning of the solutions were performed at room temperature, with a voltage of 20 kV, injection rate of 0.5 ml/h, TCD of 100 mm and the mandrel rotated with a speed of 700 rpm. The nanofibers collected on aluminum foil were stored at room temperature for the characterization tests.

2.5 Fourier transform infrared spectroscopy (FTIR)

FTIR spectra of the nanofibers were recorded by ABB (formerly Bomem) spectrometer in transmission mode. The nanofibers were rubbed and mixed with KBr powder (dried at 100°C 24 h before use), in a weight ratio of 1/100 and pressed to form a clear disc and placed in the holder in front of the beam. FTIR spectra of the samples were collected in the range 4000-400 cm^{-1} with a resolution of 2 cm^{-1} within 20 scans.

2.6 Scanning electron microscopy (SEM)

The nanofibrous mats were coated with gold without any other modification and morphology of the nanofibers was studied by SEM (AIS 2100, Seron Technology, Korea) in the magnification of 10 k. Average diameters of nanofibers were measured by Image-Pro Plus 6.0 software. For the cell cultured sample, the SEM images were obtained by the same instrument with different magnifications.

2.7 X-ray diffraction (XRD)

In order to consider the effects of electrospinning process on the crystallinity of the polymers, as-received CS and PVA powders as well as electrospun nanofibers were analyzed by a wide angle X-ray diffractometer (EQuinox 3000, INEL, France) using $\text{CuK}\alpha$ radiation source with a wavelength of $\lambda=0.154$ nm working in 40 kV and 30 mA. The intensity was recorded in the range 5° to 118° by a step size of 0.032°.

2.8 Differential scanning calorimetry (DSC)

DSC analysis was conducted under nitrogen atmosphere by Mettler Toledo Differential Scanning Calorimeter. In order to determine the effects of electrospinning on thermal behavior of the CS and PVA, these polymers were tested before and after electrospinning.

At least 5 mg of each sample was weighted, sealed in aluminum pans and heated from 25°C to 300°C by a heating rate of 10°C/min. The reference material was an empty pan. Baseline inflection point was taken as the glass-rubber transition temperature (T_g) and the endothermic peak during heating was taken as the melting point (T_m). The area of the melting peak for PVA was calculated by the instrument's software. Degree of crystallinity of the PVA was calculated using the following equation:

$$x_c = \frac{\Delta H}{\Delta H_c} \times 100 \quad (1)$$

in which ΔH is the enthalpy of melting of the sample and ΔH_c is the enthalpy of melting for 100% crystalline PVA which is assumed to be 138.6 J/g²³.

2.9 Dynamic mechanical thermal analysis (DMTA)

The DMTA experiment was performed for PVA and CS/PVA 30/70 mats. The specimens were cut from the nanofibrous mats into dimensions of 1×4 cm. The thickness of the mats, ranged from 0.01 ~ 0.03 mm, was measured by a digital micrometer. The DMTA experiments were performed using a TTDMA dynamic mechanical analyzer (Triton Technology) in tension mode at the constant frequency of 1 Hz. Temperature sweep was conducted on the samples with a rate of 5°C/min from the room temperature (25°C) up to 160°C.

2.10 Tensile strength

This test was also carried out for PVA and CS/PVA 30/70 mats. Stress-strain curves were obtained according to ASTM D882-12 (2012) using a single column universal testing machine with a 1 kN load cell (Tinius Olsen H1KS, SDL ATLAS, USA). The specimens were cut into strips of 0.5×5 cm and a gauge length of 20 mm was considered. The samples were mounted into the grips and stretched with a strain rate of 10 mm/min until breakage. Four samples of each composition were tested and averages of Young's modulus of elasticity (E), ultimate tensile strength (UTS) and elongation at break (ϵ_b) were evaluated from stress-strain curves. The data are presented as average ± standard error.

2.11 Cell behavior

Cell behavior study was performed on the CS/PVA 30/70 nanofibrous mat as the best sample of this work. Because this mat was not stable in aqueous environment, crosslinking was essential to stabilize it in aqueous media. CS/PVA nanofibers can be crosslinked by chemical^{24,25} or physical methods^{7,26}. Glutaraldehyde, glyoxal, epichlorohydrin, etc. are possible chemical crosslinkers for CS/PVA nanofibers. Heat treatment and immersion in methanol have been reported as physical methods that can stabilize CS/PVA nanofibers in aqueous media by dehydrating and increasing the crystallinity. Although the chemical methods are more efficient than physical methods in stabilizing the mats, unreacted chemical agents may remain in the structure of the nanofibers and induce cytotoxicity to the cells²⁷. In this regard, we used the physical crosslinking method by heat treatment to minimize the possible cytotoxic effects of chemical agents to the cells. The mats were heat treated at 80°C for 2 hours in order to stabilize the nanofibrous structure in water environment. Thermal treatment can increase the crystallinity of the mat and results in a more stable nanofibrous structure. For cell culture, CS/PVA 30/70 mat was sterilized by immersion in ethanol 70% for 30 minutes followed by washing with PBS for three times under sterile conditions. The L-929 cell line was obtained from national cell bank of Iran, Pasteur institute of Iran, Tehran, Iran. To study the proliferation of the cells on the surface of the mat, the L-929 cells with a density of 5×10^4 cells were seeded in a 24-well plate containing the sample at the bottom of the well supplemented by 2 ml of RPMI 1640 with 10% FBS. The plate was incubated at 37°C and 5% CO₂ for 5 days. Subsequently, the cells on the sample were fixed with 2.5% glutaraldehyde solution for 2 hours followed by washing with PBS and dehydrating with 30%, 50% 75%, 95% and absolute ethanol. The sample was dried and used for SEM observation.

3. RESULTS

3.1 Morphology of electrospun nanofibers

As mentioned before, CS is hardly electrospinnable unless by using another polymer as a guest. In this work PVA was used as the guest polymer due to its favorable interactions with CS. Concentration of PVA is an important parameter determining the electrospinnability of solutions. In this study, solutions of CS and PVA in different volume ratios were prepared. SEM micrographs of electrospun CS/PVA blends are shown in figure 2 (a-e). Uniform bead-

free morphology was observed for all samples leaving out CS/PVA 100/0 (figure 2-e) which showed beaded morphology with few fibers. The results suggest that at the total concentration of 7 wt% being constant for all samples, CS/PVA blends produce bead-free uniform nanofibers in all ratios, except the solution of pure CS. As the total concentration of polymers in the solutions were constant, formation of bead-free fibers can be related to the presence of PVA chains in the solution. In some previous works on the electrospinning of CS/PVA blends^{9,16} the CS and PVA solutions were mixed with different ratios and the total polymer concentration was changed. Thus formation of uniform nanofibers cannot only be related to incorporation of PVA and it may be due to change in the total concentration. In this work, as we only changed ratio of CS/PVA in the solutions and all other electrospinning parameters held constant, though we can conclude that presence of PVA in the CS solutions has enhanced the formation of nanofibers.

Average diameters of nanofibers are presented in Table 1. Nanofibers with diameters ranging from 172 nm (CS/PVA 70/30) to 257 nm (CS/PVA 0/100) were formed. The average diameters of nanofibers were decreased by increasing the CS content in the solutions. Similar results has been found in the previous works on electrospinning of CS/PVA^{9,16}. Formation of ultrathin fibers by electrospinning process is related to flow instabilities of the electrified jet. Repulsion forces between the charged surfaces, splits the jet and produces nano-scale fibers¹. Presence of CS in the solution, regarding its cationic polyelectrolytic nature in acidic media, increases the concentration of surface charges on the jet. Though, in samples with higher CS content, more repulsive forces were developed between the charged surfaces of the jet, leading to size depression of nanofibers.

Figure 3(a-d) shows the probability histograms of electrospun CS/PVA nanofibers. It can be seen that diameter distribution of CS/PVA 0/100 (Figure 3-a) and CS/PVA 30/70 (Figure 3-d) blend nanofibers were more uniform than nanofibers of CS/PVA 70/30 and 50/50 blends. Therefore, middle ratios of CS/PVA blends have shown more variations in the diameter of nanofibers.

3.2 Electrospinnability of the solutions

Electrospinnabilities of CS/PVA blends are presented in Table 2. It was observed that by holding the concentration of each polymer in 7 wt%, mixtures of CS/PVA solutions were

electrospinnable into uniform nanofibers in all ratios except CS/PVA 100/0 which was for the pure CS solution, leading to beaded morphology with few fibers. The reason was that in acidic media, NH_2 groups present on the CS chains, were ionized into NH_3^+ and caused strong repulsive forces between the chains in the electric field and prevented the formation of nanofibers.

The solutions with CS/PVA compositions of 30/70, 50/50 and 70/30 were electrospinnable, however, the 50/50 and 70/30 ratios had a low electrospinning throughput and very long times were needed to achieve a nanofibrous mat with acceptable thickness. In comparison, the 0/100 and 30/70 compositions had a better throughput and nanofibrous mats with appropriate thickness were achieved. According to this result, we selected the CS/PVA 30/70 as the best composition for the rest of our study and the mechanical tests and cell behavior were performed on this sample.

The viscosity of the solutions is another important parameter determining the morphology of nanofibers. Figure 4 shows the viscosities of CS/PVA solutions. Application of the electric field deforms the shape of the polymer solution to a conical shape called Taylor cone²⁸. If the electric field is strong enough, the electrostatic forces applied on the charges at the surface of the fluid, overcome the surface tension and a jet is formed. Formation of Taylor cone highly depends on the viscosity of polymer solutions. In this study, constant electric voltage (20 kV) was applied to all solutions. The Brookfield viscosity of the solutions are presented in figure 4. It can be seen that the viscosities of the CS/PVA solutions were increased by increasing the CS content in the solutions. The viscosity of CS/PVA 100/0 at the same total concentration was so high that the strength of the electric field was not sufficient for jet formation and creation of uniform continuous nanofibers.

3.3 FT-IR spectroscopy

Figure 5 shows FT-IR spectra of CS/PVA nanofibers. A number of absorption bands which are characteristic peaks of PVA and CS were observed in agreement with the literature data²⁹. OH and NH stretching vibrations were observed as a broad and intense peak around 3400 cm^{-1} for all samples. The wavenumber of this peak has shifted toward lower values by increasing the PVA content in the samples which indicates hydrogen bond formation between PVA and CS chains. The peak around 1560 cm^{-1} observed in figure 3-b-e was a

weaker N-H vibration and was depressed by increasing the PVA content in the samples. This is another indication of hydrogen bonding between PVA and CS chains.

$\text{-CH}_2\text{-}$ rocking band was detected at 720 cm^{-1} for PVA containing nanofibers (figure 5-a-d). Also the peak at 847 cm^{-1} was observed for these nanofibers and was related to C-H bond rocking. The absorbance of this peak was decreased by increasing the CS content in the samples. The peak at 1095 cm^{-1} in the PVA containing nanofibers (figure 5-a-d) is due to the vibration of C-O bonds. This peak is appeared with a shoulder and is crystallization sensitive peak of PVA. The shoulder was disappeared by increasing CS content in the samples and the shape of the peak was different for the CS/PVA 100/0. The peak at 1260 cm^{-1} is related to C-O stretching vibrations. Two peaks at 1377 and 1432 cm^{-1} were CH-OH and CH_2 symmetric bending mode vibrations of PVA.

PVA is produced by saponification of poly (vinyl acetate) and in fact, is a copolymer of vinyl alcohol and vinyl acetate repeat units. Although in this study, the PVA with 99% degree of hydrolysis was used, 1% of acetate groups remain un-hydrolyzed in the polymer. In figure 5-a-d, the peak at 1732 cm^{-1} is related to stretching vibrations of C=O groups of the remaining vinyl acetate repeat units in the PVA. The peak at 2940 cm^{-1} is related to anti-symmetric CH_2 stretching in the PVA sample. Both of these peaks were reduced by increasing the CS content in the nanofibers.

The results of FT-IR spectroscopy of the nanofibers presented in figure 5 confirm the coexistence of relevant functional groups of CS and PVA in the nanofibers. These results suggest that CS was successfully incorporated into the structure of nanofibers. In addition, FTIR spectra of nanofibers indicate the partial miscibility of CS/PVA blends. If the two polymers in the blend are not compatible, each polymer experiences the same environment as the pure polymer and the peak positions and absorbance in the spectra of the blend, are similar to that of pure components. On the other hand, if the two polymers are miscible, possible chemical interactions between the individual polymer chains, leads to the shifts in the wavenumber and bands in the spectra of the blends. In this work, the spectra of CS/PVA blends were different from that of pure polymers and the shift in the wavenumber and shape of the peaks was an evidences of hydrogen bonding interactions between the CS and PVA.

3.4 X-ray diffraction analysis

X-ray diffraction is a useful technique for analyzing the crystallinity in polymers. In order to examine the effect of the electrospinning process on the crystallinity of CS and PVA, XRD tests were performed on these polymers before and after electrospinning and the results are presented in figures 6-a and 6-b, respectively. Figure 6-a, shows the XRD patterns of as-received CS and PVA, in form of powder. XRD pattern of the PVA before electrospinning was composed of a number of crystalline peaks at 2θ of 10.8° , 15.4° , 18.7° and 19.3° and 22.2° . These peaks are an indication of different crystal types present in the structure of as-received PVA and we observed several sharp peaks which were due to high degree of crystallinity. In figure 6-a, for as-received CS, typical peaks of CS crystalline structure were seen at 2θ of 12.4° and 20.2° .

XRD patterns of nanofibers produced by blending CS with PVA in different ratios are presented in figure 6-b. For the sample CS/PVA 0/100, which consists of only PVA nanofibers, the peaks at 10.8° , 15.4° and 22.2° which were observed for the as-received PVA, were disappeared and only one sharp peak can be seen at 18.6° . This result suggests that the crystalline structure of PVA was altered after the electrospinning process. Although we expect high degree of crystallinity for fibers produced by conventional fiber spinning processes, here it was seen that the electrospinning process did not increase the crystallinity in nanofibers and has retarded the formation of crystalline microstructure, leading to a more amorphous morphology. The same result has been observed by others for PVA films and nanofibers¹⁶. The reason can be described by the concept of Deborah number. The Deborah number is defined as the ratio of characteristic time of a phenomenon to the experiment time³⁰. If Deborah number be greater than one, the time scale of the experiment is shorter than the characteristic time of the experiment, so the phenomenon is not observed. In the electrospinning process, the solution jet is highly elongated by the electric field and at the same time, the solvent is evaporated rapidly to produce nanofibers. Elongation of the jet reduces its diameter and the solvent, which is a small molecule, evaporates readily but polymer chains are macromolecules and need more time to diffuse and rearrange in the form of crystals. Solidification of the jet occurs within a time scale in the order of a second (time scale of the experiment) while the time scale for crystallization from a semi-dilute polymer solution is in the order of days (characteristic time of the

phenomenon). Here, the time scale of solvent evaporation was shorter than the time scale of crystal formation and the Deborah number was greater than one. Therefore, we did not observe the crystallization of polymer chains.

In figure 6-b, intensity of the diffraction peaks for the nanofibers were highly reduced comparing to that of as-received polymers presented in figure 6-a. The weakest crystallinity peak was observed for the sample CS/PVA 30/70. It can be said that at this blend ratio, incorporation of CS into the blend had the highest impact on the crystalline structure of both polymers and produced the most amorphous morphology. By further increasing the CS ratio in the blend, the peaks became a little stronger.

The results of X-ray diffraction also suggest partial compatibility of CS and PVA. In the blends of two immiscible polymers, each component produces its own peaks in the XRD pattern but in compatible blends, the peaks of each component are influenced by presence of the other component. Here we can observe from figure 6-b, that presence of CS in the blends has changed the diffraction patterns of PVA. One significant peak was observed at $2\theta \sim 19-20^\circ$ for each blend ratio which indicates the strong inter and intramolecular hydrogen bonding between CS and PVA chains³¹.

Another reason for the low crystallinity of CS/PVA nanofibers may be due to solvent entrapment in the polymer structure. Further discussion for this reason is explained in the DSC analysis.

3.5 Thermal characterization

DSC measurements can provide valuable data on thermal behavior and phase transitions inside materials. Glass transition and melting are two important phenomena in polymers which are affected by processing conditions and additives present in the material. Figure 7-a represents the DSC thermograms of as-received CS and PVA in form of powder before electrospinning. In the thermogram for PVA, a change in the baseline observed at 56°C is due to glass transition within the temperature range $49-63^\circ\text{C}$. The value of the glass transition temperature of the pure PVA obtained in this work was lower than that of other values reported in the literature³². This may be due to the ingress of water into the amorphous regions of PVA, acting as a plasticizer and reducing the T_g ³³. The melting peak

was observed at 190°C which started from 170°C and lasted at 200°C. Based on the peak integral normalized to the weight of the sample, the crystallinity of the as-received PVA was calculated to be 30.7%. In the thermogram of the as-received CS, the broad endothermic peak starting from room temperature and ending around 150°C is due to the presence of water molecules inside the CS structure. Three different forms of water, existing in the CS structure, start liberating at various temperature ranges. Freezing water comes out at around -5°C, tightly bound water is released in the range of ± 10 -100°C and free loosely bound water exits at about 150°C³⁴. Due to strong hydrogen bonding between its macromolecules, CS is not a thermoplastic and the degradation process occurs before the melting. Though, we did not observe any peak for the melting of the crystalline structure of CS or for the glass transition.

Figure 7-b represents the DSC thermograms of nanofibrous samples. In the thermogram of CS/PVA 0/100, a shift in the baseline was observed near 60°C which was associated to glass transition of PVA chains. The slight increase in the glass transition of the PVA nanofibers compared to that of as-received PVA may be due to the orientation of the chains occurred by the shear and elongational stresses applied by the electric field. The PVA nanofibrous mat was melted at 190°C and the degree of crystallinity of this sample was calculated to be 20.2%. This result indicates a 10% reduction in the crystallinity of the PVA nanofibers compared to that of as-received PVA and is another proof that nanofibers were highly amorphous. This result is in agreement with the XRD data. In thermograms of CS containing nanofibers (CS/PVA 30/70, 50/50 and 70/30 in figure 7-b), a broad peak was appeared from the beginning of the test, ending around 120°C. This peak was related to evaporation of bound solvents (water and acetic acid) inside the structure of CS. As we used mixture of acetic acid/water 70/30 as solvent, the evaporation was started from 50°C and finished around 120°C which was near the boiling point of acetic acid (118°C). For the nanofibers of CS/PVA 30/70 and 50/50, a small peak was observed near 60°C which is possibly related to the glass transition of PVA chains. This peak was disappeared in the nanofibers CS/PVA 70/30 possibly because the content of PVA was decreased in this sample.

In figure 7-b, for the sample CS/PVA 30/70, the melting was detectable at 191°C while melting was not observed for the nanofibers of 50/50 and 70/30. The degree of crystallinity

for the CS/PVA 30/70 nanofibers was estimated to be 8.2% which indicates very low crystallinity of this sample.

By increasing the temperature of the nanofibers in the DSC test, PVA and CS begin to degrade with different degradation behaviors. In the thermogram for the sample CS/PVA 0/100 in figure 7-b, degradation of PVA was started at the temperatures above 230°C with an endothermic peak. However, CS containing nanofibers showed a weak endothermic peak around 260°C followed by a strong exothermic peak above this temperature, reaching a maximum. The maximum temperature of the degradation peaks were observed at 280, 285 and 288°C for the nanofibers of CS/PVA 30/70, 50/50 and 70/30, respectively. The difference in the thermal degradation behavior of CS and PVA is explained as follows: by increasing the temperature above 200°C, CS macromolecules undergo crosslinking reactions followed by destruction of amino groups which is confirmed by an increase in the gel fraction³⁵ leading to appearance of the exothermic peak, while PVA mainly degrades through chain scission reactions that require energy and show endothermic peak.

3.6 Tensile properties

As mentioned before, due to the low electrospinning throughput of the solutions containing more than 30% CS, we selected the PVA and CS/PVA 30/70 nanofibrous mats for tensile strength test. Figure 8 shows the tensile behavior of PVA and CS/PVA 30/70 nanofibrous mats. Ductile behavior was observed for the PVA mats while the CS/PVA 30/70 mat showed a brittle behavior. The PVA mat (solid line) had a much higher tensile strength and elongation at break compared to the CS/PVA 30/70 (dotted line) mat. The tensile strength was reduced for the CS/PVA 30/70 nanofibrous mat and the sample was broken at a lower extension and stress.

The results of ultimate tensile strength, Young's modulus, elongation at break as well as toughness of the mats are summarized in Table 3. The Young's modulus, ultimate tensile strength, elongation at break and toughness were reduced by incorporation of CS.

3.7 DMTA results

DMTA experiments can provide valuable data on phase transitions and viscoelastic behavior of polymers and polymer blends. Figure 9 presents the results of DMTA test for the

nanofibrous mats of PVA and CS/PVA 30/70. From figure 9-a, it was seen that when the temperature was lower than 80°C, the storage modulus for the PVA mat was higher than CS/PVA 30/70 mat which was in accordance to the tensile strength data. CS chains are stiffer than PVA chains and it was expected that the storage modulus of CS/PVA mat be higher than the PVA mat. Other researchers observed that for the CS/PVA films the storage modulus was increased by incorporation of CS³⁶. However, in this work, the storage modulus was decreased by incorporation of CS in the nanofibrous mat. The reason for this reduction may be due to the reduced crystallinity, presented by the XRD data. Also, it may be due to the bound solvents in the structure of nanofibers which contain CS. As the DSC curves of figure 7-b show, a wide endothermic peak due to evaporation of bound solvent (acetic acid 70%) was observed for the sample CS/PVA 30/70 while this peak was absent for the PVA mat. This amount of bound solvent, acted as a plasticizer for the CS/PVA 30/70 mat and reduces its storage modulus compared to that of PVA mat. Additionally, for the PVA mat, the storage modulus was decreased by increasing the temperature above the room temperature while for the CS/PVA 30/70, the storage modulus was slightly increased by increasing the temperature up to 60°C and then decreased. It was also due to the evaporation of bound solvent between the CS chains.

The loss factor, $\tan \delta$, is the ratio of loss modulus to the storage modulus which can be attributed to the relaxation processes in polymers and polymer blends. For the PVA mat, the intensive relaxation peak with a maximum at 86°C (figure 9-b) attributed to the glass transition temperature of PVA. The DMTA experiments usually give higher values for the glass transition temperature in comparing to the DSC method. This result was also observed in other works and a glass transition around 95°C was reported³⁷. In the case of CS/PVA 30/70, a relaxation peak in the $\tan \delta$ around 45°C was observed which is related to the structural reorganization of CS macromolecules due to increased mobility of water molecules, volume expansion and change in the hydrogen bond strength³⁴. By further increasing the temperature, the $\tan \delta$ reached a maximum at around 110°C with a broader peak compared to the PVA curve, due to close placement of relaxations of PVA and CS. The maximum value for $\tan \delta$ was reduced from 0.45 for the PVA mat to 0.28 for the CS/PVA 30/70 mat and also the values were diminished for the latter, over the temperature range of the experiment.

3.8 Cell attachment and proliferation

Figure 10 (a,b) represents the SEM micrographs of attachment and proliferation of L-929 fibroblast cells on the surface of thermally stabilized CS/PVA 30/70 nanofibrous mat after 5 days of incubation. From figure 10-a, it can be seen that the cells were appropriately attached to the surface of the nanofibrous mat and achieved extended morphology. Figure 10-b shows that the cells were connected together by extending their filopodias. It is also notable that the nanofibrous mat retained its morphology during the experiment period, although the nanofibers were merged together due to the swelling in aqueous media. Results of cell attachment study showed that CS/PVA 30/70 nanofibrous mat was able to support the attachment and proliferation of fibroblast cells without any cytotoxic effect.

4. Discussion

CS/PVA blends in form of nanofibers and films have been extensively studied in the literature^{6,14,38-41}. In this study, we observed that crystallinity of the nanofibrous mats were less than the as-received powders, which was confirmed by the DSC and XRD data. Other researchers have also reported the retardation of crystallization by the electrospinning process^{16,42-44}. In this regards, our results are in accordance with the previous publications. In the electrospinning process, the high voltage electric field, applies a significant elongational strain to the solution which results in the stretching of polymer chains. In the later stages of the electrospinning process, rapid solidification of the stretched polymer chains is required to form the nanofibers. Therefore, this rapid solidification hinders the formation of the crystalline structure and leads to a lower crystallinity for the nanofibers compared to the powders or films.

Tensile strength of CS/PVA blend films with different ratios have been reported in the previous works. The effect of PVA on tensile strength of the films have been the subject of controversy. Park and coworkers⁴⁵ reported that the tensile strength of the films were increased by increasing the PVA content in the blends, while others³⁸ observed that tensile strengths of the films were reduced by incorporation of PVA. In this study, the tensile strength of the nanofibrous mats were reduced by incorporation of CS in the blend. According to XRD data (figure 6), for the nanofibrous mats of CS/PVA 0/100 and 30/70, we observed that the crystallinity of the CS/PVA 30/70 was reduced compared to the PVA mat.

Therefore, the reduction in the tensile strength can be attributed to the reduced crystallinity of the CS/PVA 30/70 nanofibers comparing to the PVA nanofibers.

It is well known that DMTA can be used to investigate miscibility of polymer blends. In a miscible blend, the T_g is represented by a single peak with an intermediate value between the T_g of the pure components. As mentioned, from Figure 9-b it is seen that the relaxation in CS/PVA 30/70 blend has occurred with a broader $\tan \delta$ peak. It indicates that presence of CS chains has altered the onset of molecular motions in the PVA chains. This comes from the strong hydrogen bonding between PVA and CS chains and is another indication of partial compatibility between CS and PVA, in agreement with DSC and FTIR data. The broadness of the peak in CS/PVA 30/70 may be due to differences in the chain structure of PVA and CS.

PVA chains are formed by repetition of vinyl alcohol while CS has a saccharide backbone. CS chains have a limited molecular motion within temperature range of the experiment.

Therefore, it is expected that addition of CS chains between PVA chains, increase the glass transition temperature of the blend. However, in the $\tan \delta$ curve of the nanofibrous mats of CS/PVA in this study, a single T_g peak was not observed as reported by previous works on the CS/PVA films³⁷.

CS and PVA are both hydrophilic polymers and the nanofibers produced from their blends dissolve quickly in the aqueous environment. Though, chemical or physical treatment is required to stabilize the mats in water and perform the cell culture. When the mats are fixed by a crosslinking agent (e.g. glutaraldehyde), time-consuming washing steps are needed to remove the unreacted chemical, because they have the potential to induce cytotoxicity to the cells⁴⁶. Physical methods such as thermal treatment^{47,48} or soaking the mats in methanol²⁶ were also used in the literature. As methanol is also a cytotoxic chemical, we selected the thermal treatment as a safe method to stabilize the mats for cell culture. Thermal treatment of CS/PVA nanofibers, dehydrates the mats and increases the crystallinity⁴⁹ which results in the stabilization of the mat in the aqueous environment⁵⁰. Results of the cell attachment study in this work, approved that the heat stabilized CS/PVA nanofibrous mats were able to support attachment and proliferation of L929 fibroblast cells without any adverse effect.

5. CONCLUSIONS

CS/PVA nanofibers with acceptable electrospinning throughput can be prepared from CS/PVA 30/70 solutions using acetic acid 70% as solvent. FTIR spectra of nanofibers demonstrated the partial compatibility of CS/PVA blends through favored chemical interactions and hydrogen bonding. The XRD data also confirmed the partial miscibility of the blends. Comparing the XRD and DSC data for the as-received CS and PVA with that of nanofibers, it was observed that electrospinning process has lowered the crystallinity of the polymers. In accordance to the tensile strength results, the storage modulus of the CS/PVA 30/70 mat was reduced in comparison to the PVA mat. The $\tan \delta$ curves of the PVA mat showed an intensive relaxation in the glass transition temperature while a broad relaxation was observed for CS/PVA 30/70. The CS/PVA 30/70 mat has been able to support the attachment and proliferation of L929 fibroblast cells.

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Figure legends

FIGURE 1 Set up used for electrospinning of nanofibers.

FIGURE 2 SEM micrograph of nanofibers electrospun from CS/PVA solutions in different ratios: (a) 0/100, (b) 30/70, (c) 50/50, (d) 70/30 and (e) 100/0.

FIGURE 3 Probability histogram of CS/PVA nanofiber diameters with different ratios: a) 0/100, b) 30/70, c) 50/50 and d) 70/30.

FIGURE 4 Viscosities of CS/PVA blend solutions measured by Brookfield viscometer at 60 rpm.

FIGURE 5 FT-IR spectra of CS/PVA nanofibers with different ratios: (a) 0/100, (b) 30/70, (c) 50/50, (d) 70/30 and (e) pure CS.

FIGURE 6 XRD patterns of (a) CS and PVA before electrospinning and (b) electrospun mats with different CS/PVA ratios.

FIGURE 7 DSC thermograms of (a) as-received CS and PVA and (b) electrospun nanofibers with different ratios of CS/PVA at the heating rate of 10°C/min under nitrogen atmosphere.

FIGURE 8 Tensile behavior of PVA and CS/PVA 30/70 nanofibrous mats measured at the extension rate of 10 mm/min.

FIGURE 9 Results of DMTA experiment for PVA and CS/PVA 30/70 nanofibrous mat: (a) storage modulus and (b) $\tan \delta$ measured at the rate of 5°C/min.

FIGURE 10 Morphology of L-929 fibroblast cells cultured on the CS/PVA 30/70 nanofibrous mat after 5 days.

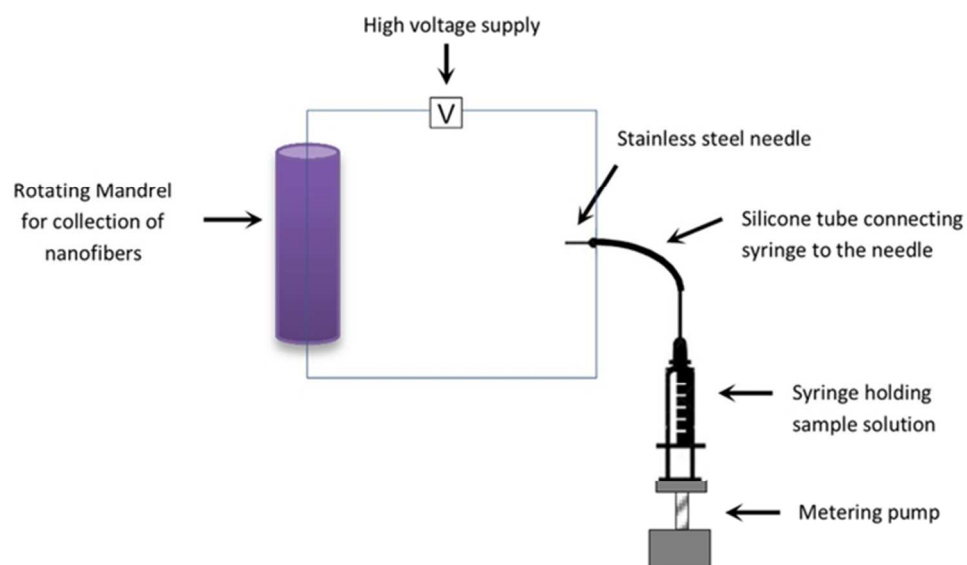


Figure 1-Set up used for electrospinning of nanofibers.
59x34mm (300 x 300 DPI)

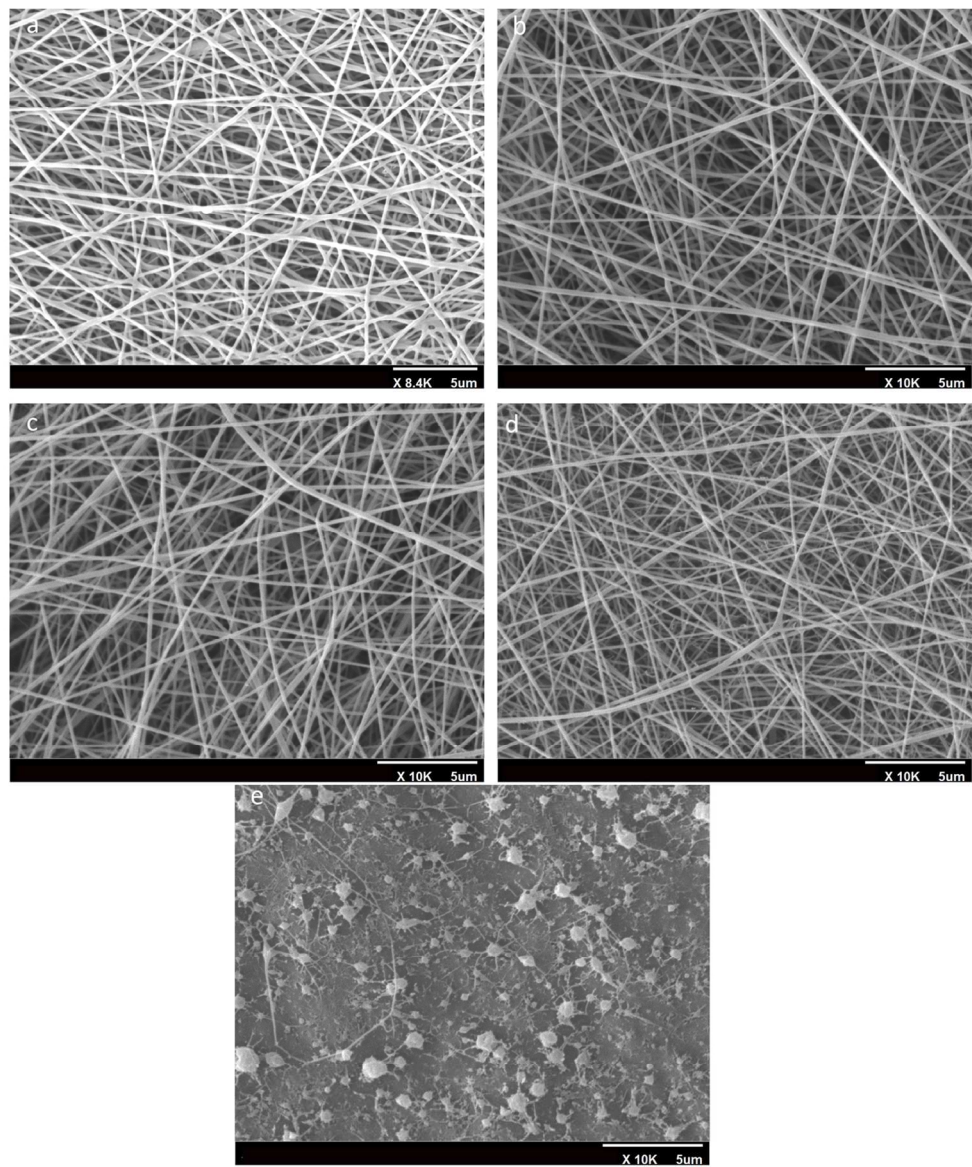


Figure 2-SEM micrograph of nanofibers electrospun from CS/PVA solutions in different ratios: (a) 0/100, (b) 30/70, (c) 50/50, (d) 70/30 and (e) 100/0.
127x148mm (300 x 300 DPI)

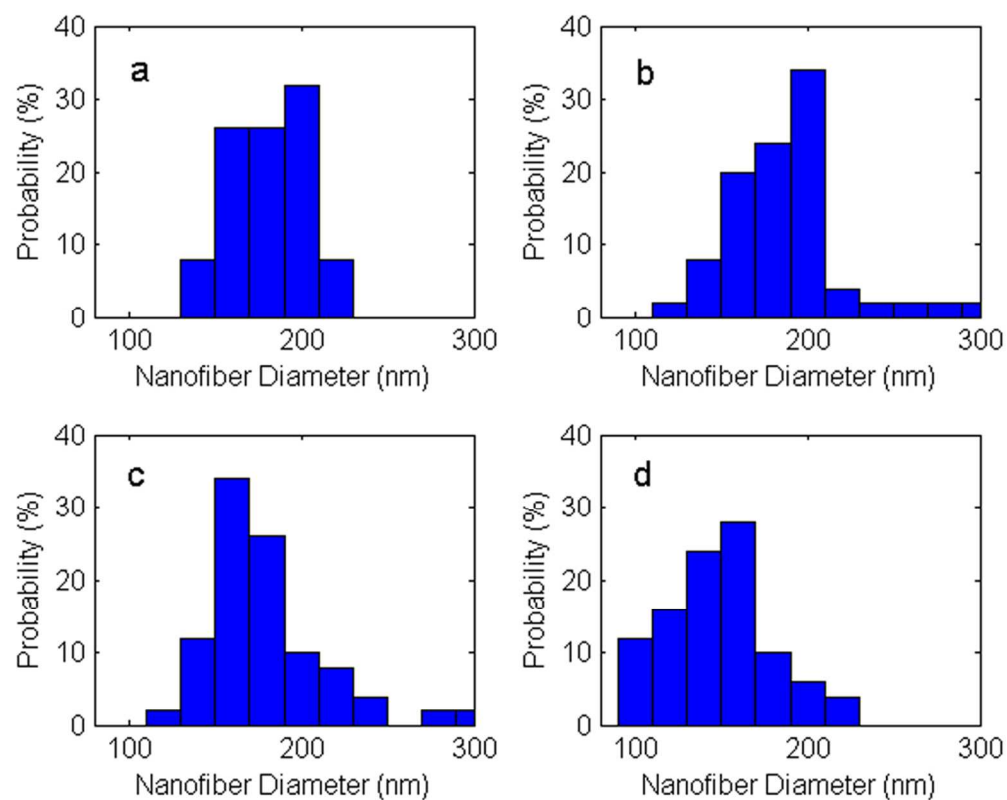


Figure 3-Probability histogram of CS/PVA nanofiber diameters with different ratios: a) 0/100, b) 30/70, c) 50/50 and d) 70/30.
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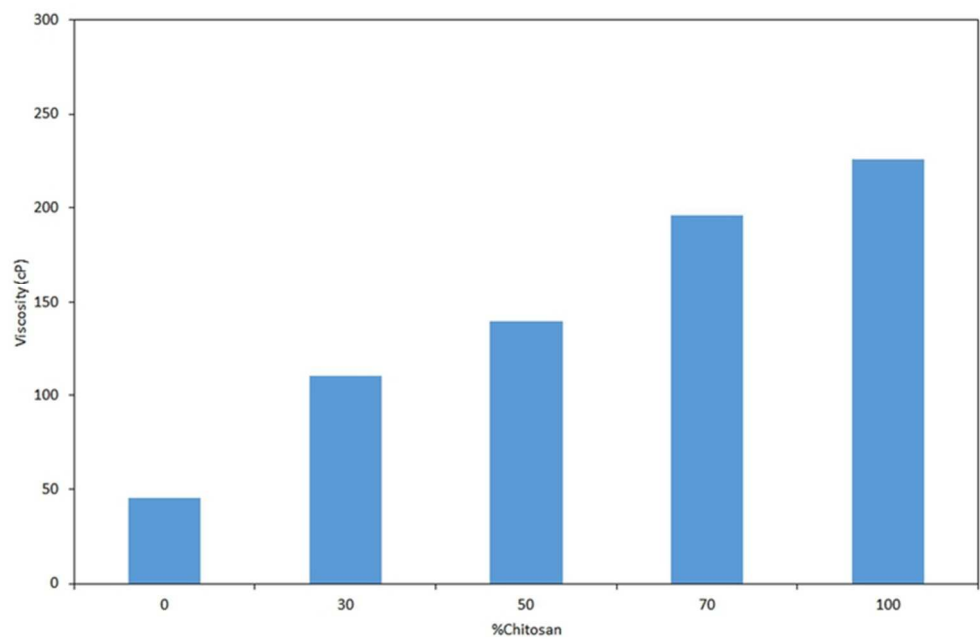


Figure 4-Viscosities of CS/PVA blend solutions measured by Brookfield viscometer at 60 rpm.
66x43mm (300 x 300 DPI)

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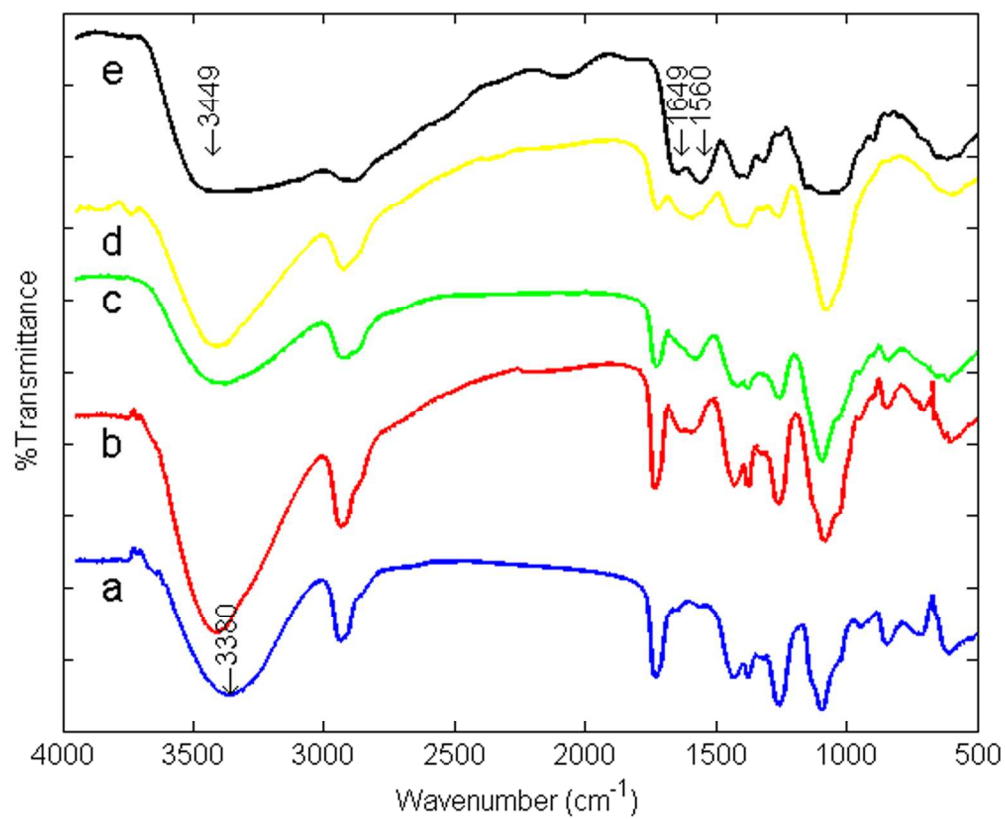


Figure 5-FT-IR spectra of CS/PVA nanofibers with different ratios: (a) 0/100, (b) 30/70, (c) 50/50, (d) 70/30 and (e) pure CS.
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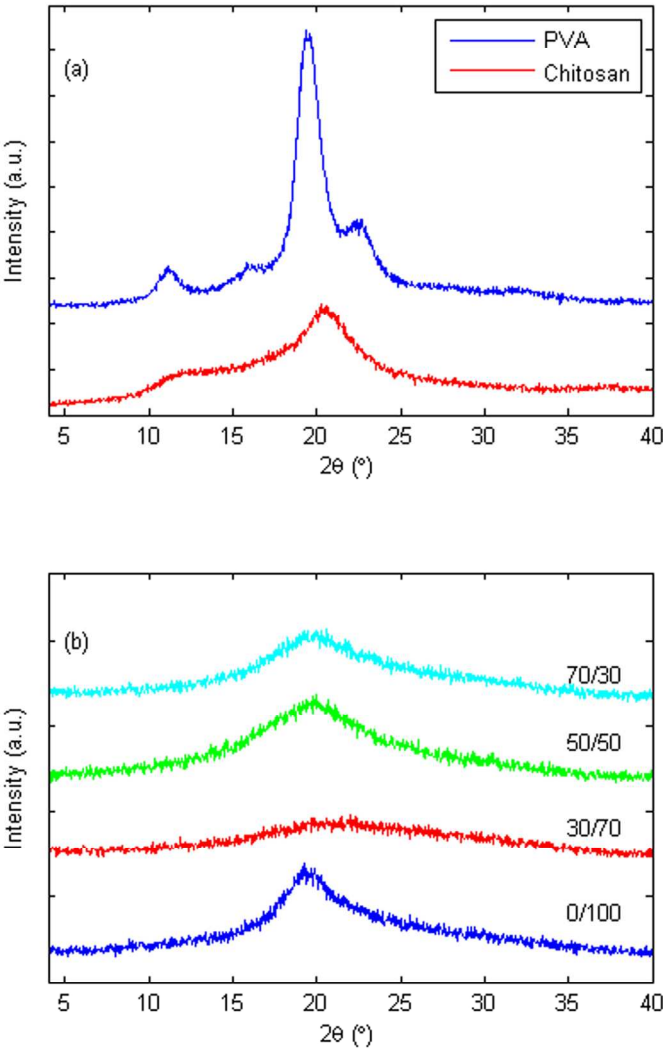


Figure 6-XRD patterns of (a) CS and PVA before electrospinning and (b) electrospun mats with different CS/PVA ratios.
156x242mm (300 x 300 DPI)

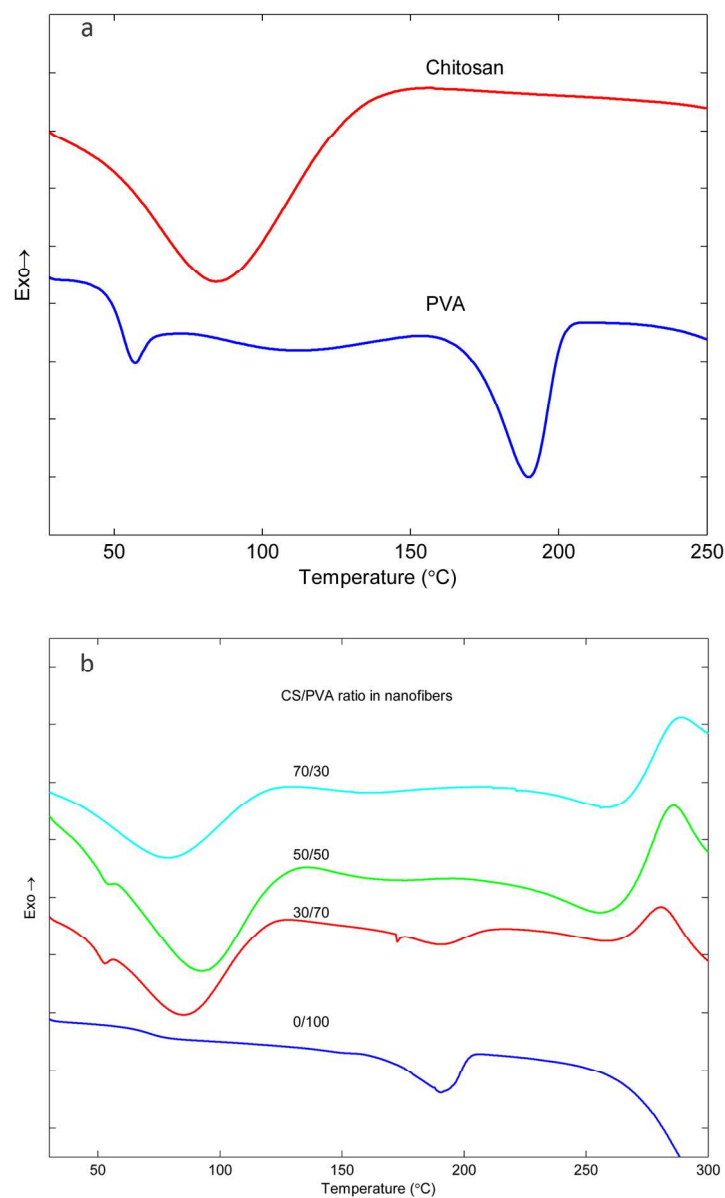


Figure 7-DSC thermograms of (a) as-received CS and PVA and (b) electrospun nanofibers with different ratios of CS/PVA at the heating rate of 10 °C/min under nitrogen atmosphere.
127x207mm (300 x 300 DPI)

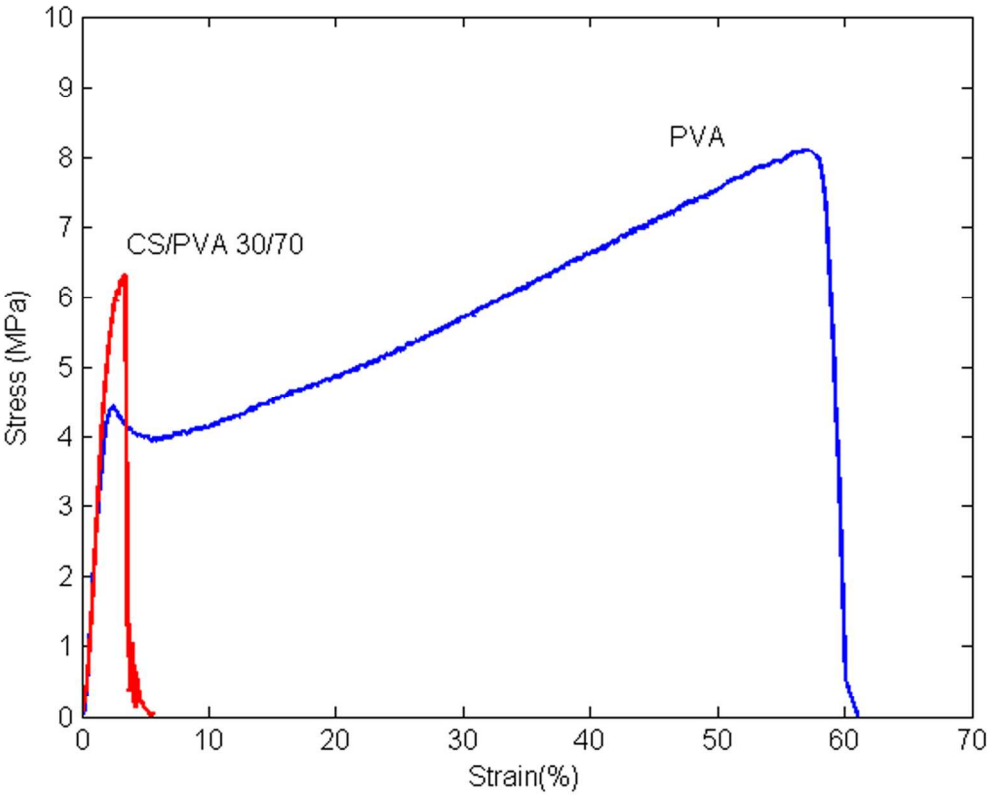


Figure 8-Tensile behavior of PVA and CS/PVA 30/70 nanofibrous mats measured at the extension rate of 10 mm/min.
118x94mm (300 x 300 DPI)

Accep

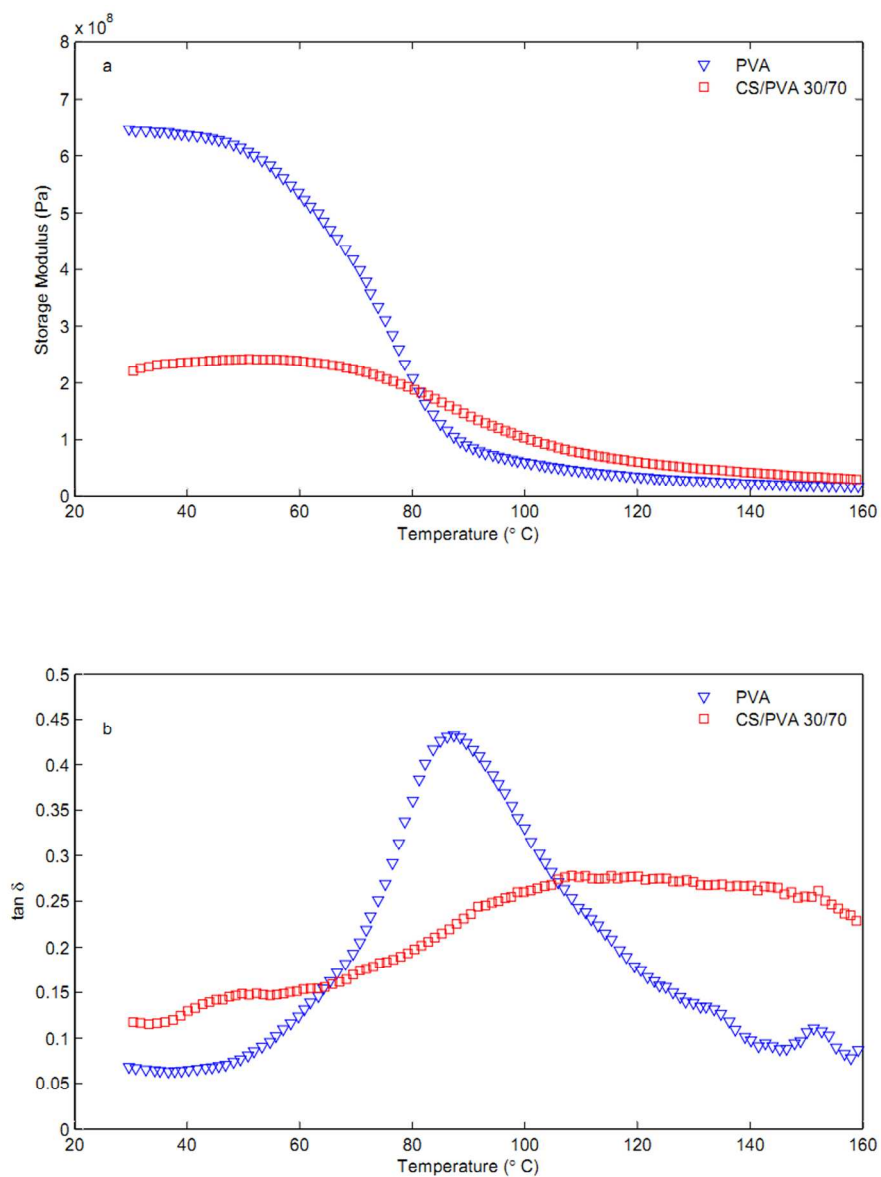


Figure 9-Results of DMTA experiment for PVA and CS/PVA 30/70 nanofibrous mat: (a) storage modulus and (b) $\tan \delta$ measured at the rate of 5°C/min.
186x254mm (300 x 300 DPI)

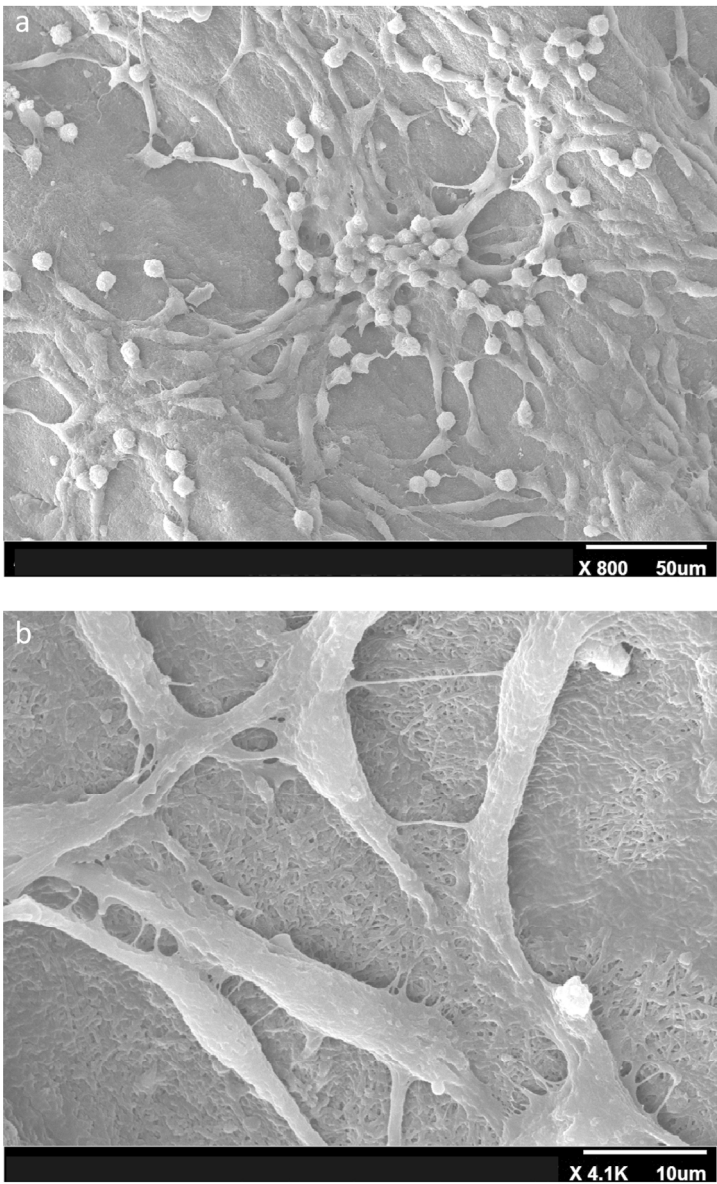


Figure 10-Morphology of L-929 fibroblast cells cultured on the CS/PVA 30/70 nanofibrous mat after 5 days.
101x167mm (300 x 300 DPI)

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CS/PVA ratio	Average Diameter (nm)
0/100	257 ±63
30/70	223 ±50
50/50	182 ±54
70/30	172 ±60

Table 1 Average diameters of nanofibers of CS/PVA with different ratios measured from the SEM micrographs.

CS/PVA ratio	Electrospinnability
0/100	possible
30/70	possible
50/50	Possible (low throughput)
70/30	Possible (low throughput)
100/0	not possible

Table 2 Electrospinnability of the CS/PVA blends with different ratios (Voltage 20 kV; TCD 200 mm; rate 0.5 ml/h)

sample	E (MPa)	UTS (MPa)	ϵ_b (%)	Toughness (MJ/m ³)
PVA	280 ±25	7.4 ±0.37	50.2 ±5	2.70 ± 0.35
CS/PVA 30/70	190±37	5.26 ±0.53	4.5 ±1	0.153 ±0.05

Table 3 Tensile properties of CS/PVA nanofibers (E: Young's modulus, UTS: Ultimate Tensile Strength, ϵ_b : elongation at break).